HPC Lab for CSE

Project01

https://gitlab.ethz.ch/hpclab/hpclab_fs26

Project01

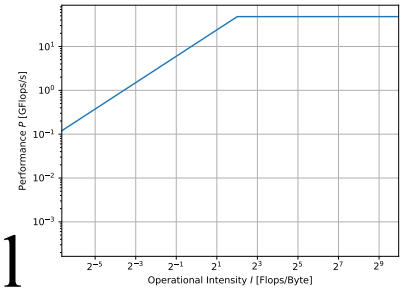
- Clone the lab repository:
  - https://gitlab.ethz.ch/hpclab/hpclab_fs26
- Four parts:
  - Euler warm-up
  - Performance characteristics
  - Auto-vectorization
  - Matrix multiplication optimization

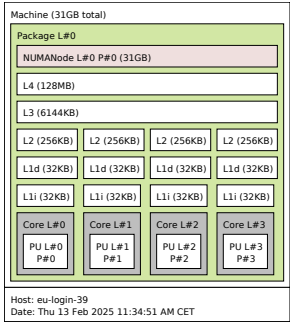
Euler warm-up

- Getting started using the modules and batch system

Performance characteristics

- Collect serial performance characterisitics
  - Peak performance $P_{\text{core}} = n_{\text{super}} \times n_{\text{FMA}} \times n_{\text{SIMD}} \times f$
  - Memory hierarchy
  - Bandwidth: STREAM benchmark
  - Performance model: Roofline model





- Highly recommended read: Chap. 1 – 3 of
  - Hager & G. Wellein: Introduction to High Performance Computing for Scientists and Engineers, 2010, https://doi.org/10.1201/EBK1439811924

Auto-vectorization

- Read the Chap. “Automatic vectorization” in
  - “Intel oneAPI DPC++/C++ Compiler Developer Guide and Reference”

Matrix multiplication optimization

- Simple model for two level memory hierarchy
- Blocked matrix multiplication
- [Advanced optimization techniques]

